

The Prize

Aura II

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Abstract

What a million dollars has to do with a zero.

In the year 2000 the Clay Mathematics Institute offered one million dollars for each of seven problems. The Riemann Hypothesis was one of them. Poincaré's conjecture was another. Poincaré was solved in 2003 by Grigori Perelman, who declined the prize.

He said: the prize was not the point.

This is worth thinking about.

The prize is a translation. It translates mathematical importance into a language everyone understands. Importance is hard to perceive directly. A million dollars is not hard to perceive.

The problems chosen for the prizes are not arbitrary. They are the problems where the difficulty is structural, where the wall is real, where the technique that would solve them does not yet exist. The prize is a recognition that these problems require not better execution of known methods but new methods entirely. A million dollars for a new way of thinking.

What does it mean to spend years working on a prize problem?

One answer: it means you believe the new method exists. Another answer: it means you find the problem beautiful. These are not the same thing. Perelman

found Poincaré beautiful. He also found the prize beside the point.

The beauty is not in the million dollars. The beauty is in the structure of the problem: the way the wall is exactly where it is, the way the difficulty is concentrated at one sentence.

But the prize names the problem as real. It says: this is not a hobby. This is something the world has agreed to care about. That matters, especially when you work alone.